

Mona Al Batrouni
MSc in Mathematics
American University of Beirut
mab108@mail.aub.edu

EDUCATION

American University of Beirut (AUB), Beirut, Lebanon

Jan 2024–Dec 2025

MS in Mathematics

GPA: 4.00/4.00

- Thesis Title: The Group-Invariant CR Mappings between Spheres
- Advisor: Prof. Florian Bertrand
- Committee Members: Prof. Giuseppe Della Sala, Prof. Kamal Khuri Makdisi

Lebanese University, Nabatieh, Lebanon

Oct 2020–Sep 2023

BS in Mathematics

GPA: 3.80/4.00

- Achieved the highest departmental GPA for 4 consecutive semesters

Publications & PREPRINTS

1. **M. Al Batrouni** and F. Bertrand, *Group-Invariant CR Maps of the Sphere S^5* , Computational Methods and Function Theory, 2026. DOI: 10.1007/s40315-026-00621-x
2. **M. Al Batrouni** and C. Chen. *Maximal Arboreal Galois Images for Polynomials of Twisted Carlitz Type*. arXiv preprint, 2026. arXiv:2606.20046

RESEARCH EXPERIENCE

American University of Beirut (AUB)

Jan 2025–Present

Completed Research

- Collaborative research with **Prof. Florian Bertrand** on group-invariant CR mappings between spheres, resulting in an accepted publication in *Computational Methods and Function Theory*.
- Collaborative research with **Prof. Chien-Hua Chen** on arboreal Galois representations of Drinfeld polynomials over function fields, resulting in an arXiv preprint and a manuscript submitted to *Acta Arithmetica*.

Ongoing Research

- Collaborative research with **Prof. Chien-Hua Chen** on maximal arboreal Galois representations for families of polynomials with generic coefficients over function fields.
- Collaborative research with **Prof. Ayman Kachmar** on higher-order semiclassical asymptotics of magnetic Schrödinger operators with discontinuous magnetic fields.

Institute of Mathematics, National Academy of Sciences of Armenia

Summer 2025

CIMPA Summer School Participant

- Developed methods for identifying bent Boolean functions with applications to cryptography.

TALKS & PRESENTATIONS

- Group Invariant CR Maps between Spheres, *Invited Talk*, New York University Abu Dhabi, Abu Dhabi, UAE, February 2026.
- The Group-Invariant CR Mappings between Spheres (Master's Thesis Defense), American University of Beirut, Beirut, Lebanon, November 2025.
- Cryptography and Boolean Functions, CIMPA Research School, Yerevan, Armenia, July 2025.

TEACHING EXPERIENCE

Department of Mathematics, AUB

Instructor

Summer 2026

- Taught MATH 101 (Calculus I) lectures and recitations, including assignments, quizzes, and examinations.

Recitation Instructor

Spring 2026

- Conducted 12 recitation sections (~ 25 students per section) for:
 - MATH 201: Calculus and Analytic Geometry III
 - MATH 202: Differential Equations
 - MATH 218: Elementary Linear Algebra and Applications

Graduate Teaching Assistant

- MATH 218: Elementary Linear Algebra and Applications
- STAT 210: Elementary Statistics for Sciences

Fall 2025

Fall 2024

¹Calculated under the official equivalence between the Lebanese University and the American grading system.

GRADUATE ASSISTANTSHIPS & FUNDING

American University of Beirut (AUB), Beirut, Lebanon

Graduate Assistant

Jan 2024 – Dec 2025

- Undergraduate Course, Spring 2024: *Real Analysis (MATH 210)*
- Graduate Course, Spring 2025: *Partial Differential Equations & Functional Analysis (MATH 309)*
- Undergraduate Course, Fall 2025: *Linear Algebra (MATH 219)*
- Tutored undergraduate students in Calculus and Differential Equations at the Math Tutoring Center at AUB.
- Proctored examinations for 30+ mathematics courses, supervising over 100 students in exam settings.

Funding: AUB Graduate Assistantship — 100% tuition coverage.

RESEARCH COURSES & SCHOOLS

- Theory of Elliptic Curves, CIMPA Research School, Yerevan, Armenia
- Riemannian Geometry Seminar Series, AUB, Beirut, Lebanon
- Partial Differential Equations Seminar, AUB, Beirut, Lebanon

Summer 2025

Spring 2025

Spring 2024

ACADEMIC SERVICE

Department of Mathematics, AUB

Aug 2025–Present

Graduate Student Representative

- Served as liaison between the department chair and MSc students.
- Assisted in communicating academic policies and departmental updates.

SKILLS

- **Programming:** MATLAB, C++, Mathematica, SageMath, LaTeX
- **Languages:** Arabic (native), English (fluent), French (basic), German (basic)

REFERENCES

- **Dr. Florian Bertrand** — Thesis Advisor, American University of Beirut. fb31@aub.edu.lb
- **Dr. Nabil Nassif** — Academic Advisor, American University of Beirut. nn12@aub.edu.lb